# Multi Area OSPFv3 Configuration in a SOHO based topology with the incorporation of Multi-Layer Switches

Krish Thapaliya

## Purpose

This guide/analysis is meant for personnel who wish to configure OSPFv3 to enable dynamic routing, create a highly efficient network, and reduce the manual configurations that a network administrator must perform to set up and maintain a network. Also, the configuration and explanation of the various related topics and devices provide a refresher for any network administrator who is trying to create an organized network through the various options that OSPF offers such as Areas, Link State Database, or Dynamic Routing.

## Background

Open Shortest Path First (OSPF) is a routing protocol for IPV4 and IPV6 networks. It uses a link state advertisement (LSA) routing algorithm and operates within a single autonomous system (AS). The LSA algorithm provides each router with a complete topology of the network by sending and receiving LSAs that describe a router's neighbors to all other routers. Each router then independently uses the shortest-path algorithm on this complete network topology to calculate the least-cost path to every destination and build its forwarding table. This process creates a uniform view of the network for all routers, leading to reliable routing and fast convergence.

A single autonomous system (AS) is a group of IP networks and routers that are centrally controlled by a single administrative entity that adheres to a single, well-defined routing policy. The best-known administrative entities are Internet Service Providers (ISP) like AT&T, Verizon, China Mobile etc. For unique policies and configurations these entities will use the unique Autonomous System Number (ASN) that is assigned to every Autonomous System (AS) by a regional internet registry, which is then used to identify it globally on the internet.

Focusing back on OSPF, the routing protocol was originally designed in the 1980s, with version 2 that we currently use for IPV4 being defined in RFC 2328 in 1998 and version 3 we use for IPV6 in RFC 5340 in 2008. The phrase RFC followed by a number, refers to the specific formal document that the Internet Engineering Task Force (IETF) creates, including the complete details, rules, and mechanism of this version of the routing protocol.

The IETF is a volunteer organization that develops open standards for the internet such as IP and TCP and works to make the Internet work better by producing high-quality, relevant technical and engineering documents through a process of collaboration and research among its members including people from all over the tech industry.

OSPF uses a unique system of organization called areas which divides a network into routing areas to simplify administration and optimize traffic and resource utilization. Areas are identified by 32 bit-numbers that can be expressed as an integer in the range of 0 to 4,294,967,295. The proper format for assigning area's a number is to make area 0 represent the core/backbone area of an OSPF network. Identifications of other areas may be chosen at will, but administrators often select the IP address of a main router in an area as the area identifier. Each additional area must have a connection with the OSPF backbone which is typically maintained by a router that has interfaces in more than 1 area known as an area border router (ABR). An ABR maintains separate Link-State Databases for each area it serves and maintains summarized routers for all areas in the network.

There are 2 types of area configuration for OSPF: Single Area OSPF and Multi Area OSPF. A Single Area OSPF network is a basic OSPF implementation where all routers are configured to belong to a single OSPF area which according to proper format should be Area 0. In Single Area OSPF, all routers have a complete view of the entire network's topology, forming a single Link State Database (LSDB) within that area. This specific type of OSPF is designed for small networks usually found in homes and any change in the topology of the network will require every router to re-calculate the optimal path, which is resource intensive. For Single Area OSPF to be activated the protocol must be manually configured, so if OSPF is not enabled the network is not using either Single Area OSPF or Multi Area OSPF.

The type of OSPF we will be focusing on and implementing in this analysis is Multi Area OSPF. Multi Area OSPF is a hierarchical network design that divides a large OSPF domain into smaller, more manageable sections called areas. By doing this, it addresses the scaling limitations of Single Area OSPF networks, such as large routing tables, extensive Link States Databases (LSDB) and frequent Shortest Path First (SPF) algorithm recalculations taking up large amounts of resources. Just like in Single Area OSPF, Area 0 should be assigned to the backbone/core. However unique to Multi Area OSPF, all other areas must have a direct or virtual connection to Area 0 which causes all traffic between non-backbone areas to pass through the backbone.

The Area Border Routers (ABRs) are routers with interfaces in 2 or more areas, with at least one of the interfaces in Area 0. An ABR maintains a separate LSDB for each area it is connected to. This is the key to multi-area OSPF's since it keeps the area topology information contained within the area's routers. The ABRs can even be configured to summarize routes between areas to bundle multiple network addresses from outside the area as a single summary rote, significantly shrinking routing tables in other areas reducing potential costs for processing. Even if there is a topology change in the network only the area where the change has occurred will trigger an SPF recalculation making the impact from a network wide to a local change. This reduces the SPF calculation, improves network stability, and has more efficient resource usage.

Besides Single Area vs. Multi Area, OSPF also has 2 main versions, OSPFv2 and OSPFv3. OSPFv2 is designed for IPV4 networks to find the shortest path to destinations within an autonomous system. This version of OSPF runs on IPV4 subnets and includes the network mask in its configuration. The packet headers that move OSPFv2 are 24 bytes, while using mechanisms like MD5 hashing for authentication and integrity. It even uses 7 types of Link State Advertisements (LSAs) to describe the network topology but if there is a change done to the topology a new type 1 LSA or even type 2 LSA will be flooded throughout the entire network.

OSPFv3 is a more efficient version of OSPF that was primarily designed for IPV6 but can also be configured for IPV4 networks, incorporating several improvements to support the new protocol's characteristics. There is no network addressing like OSPFv2 as OSPFv3 operates on links rather than subnets and uses IPV6 link-local addresses for communication cutting down on potential costs. The header for OSPFv3 are a smaller 16 bytes and relies on the IPsec protocol for authentication and encryption, removing these functions from the OSPFv3 packet header creating a more efficient network.

Instead of 7 LSAs, OSPFv3 supports 9 LSA types, with redesigned LSAs that separate topology information from addressing, improving efficiency and flexibility. In terms of commands, OSPFv3 does not use the network command that OSPFv2 uses and instead employs interface commands to enable OSPFv3 on an interface. Finally, OSPFv3 separates the SPF tree and prefixes, increasing the efficiency of the network as when a topology change occurs, the router only has to flood an updated link LSA and intra area prefix LSA without any of the routers having to run SPF in the case when the link local address changes.

One of key devices used in the topology of this analysis is the Multi-Layer Switch which is significantly different from normal Layer 2 Switches that are more common to see in networks. A Layer 2 Switch operates at the Data Link (OSI Layer 2) and forwards data based on MAC addresses, suitable for LAN communication within a single network segment or VLAN.

In contrast, the Multi-Layer Switch, also known as a Layer 3 Switch combines the functionalities of a Layer 2 switch and a router, operating at both the Data Link and Network Layers. This allows the forwarding of both MAC and IP Addresses, capability to route between subnets or VLANs, increased capacity, security and traffic management for complex networks.

The Layer 2 Switch in summary can forward data frames based on the destination MAC Address; Operates within only a single network segment or VLAN; Does not have routing capabilities between different VLANS or subnets; Ideal for connecting devices within a single LAN to reduce traffic; and generally, costs less than a Multi-Layer Switch.

The Multi-Layer Switch in summary integrates Layer 2 switching with Layer 3 routing being able to use IP Addresses for forwarding traffic between different networks and subnets, in addition to MAC Addresses for local delivery; Capable of routing traffic between different networks, subnets and networks; Provides static and dynamic routing functions allowing to act as a route within the network; Standard use for complex enterprise networks, data centers, enhanced security, and advanced traffic management; Requires more complex configuration due to its advanced routing and Layer 3 features; and higher cost than Layer 2 Switches.

## Lab Summary

In this analysis/guide, we set up a basic topology including devices like Routers and Multi-Layer Switches with a focus on setting up OSPFv3 for IPV4 and IPV6 to simulate how a network would be configured and organized in an enterprise environment. The Configurations included example IP Addresses that saved the most amount of space and other unique configurations offered in OSPF such as Router IDs, Areas and Link State Advertisements which are essential configurations that any network administrator should be familiar with.

## Lab Commands

*Switch(config)#sdm prefer dual-ipv4-and-ipv6 default*

Sets the Switch Database Management (SDM) template to a template that supports both IPv4 and IPv6. This command will only be implemented and applied to the switch after the user uses the *reload* command. This must be entered on a Layer 2 switch before IPv6 services will function. Note: This command does not need to be entered into a Multilayer Switch.

*Switch(config)#ip routing*

*Switch(config)#ipv6 unicast-routing*

Enables the use of static routes and all routing protocols for IPv4 and IPv6 as well as the forwarding of packets. These commands must be entered on a switch before IP addresses can be configured for devices to function properly.

*Switch(config)#router ospf <process-id>*

Configures a new OSPF for IPV4 instance that contains all the subsequent OSPFv2 configuration. The process ID can be a number either from 1 to 65,535, that uniquely identifies the OSPFv2 process on that local switch and does not need to match the process ID of neighboring routers.

*Switch(config-router)#router-id <ip-address>*

The router ID is a 32-bit identifier written in IPV4 format that can be manually configured to the switch that is a fundamental element of OSPFv2 and OSPFv3. It is used in IPV4 and IPV6 networks to identify the switch in all OSPFv2 and OSPV3 exchanges and when viewing neighbor relationships.

Switch(config)#*ipv6 router ospf <process-id> ipv6 area <area-id>*

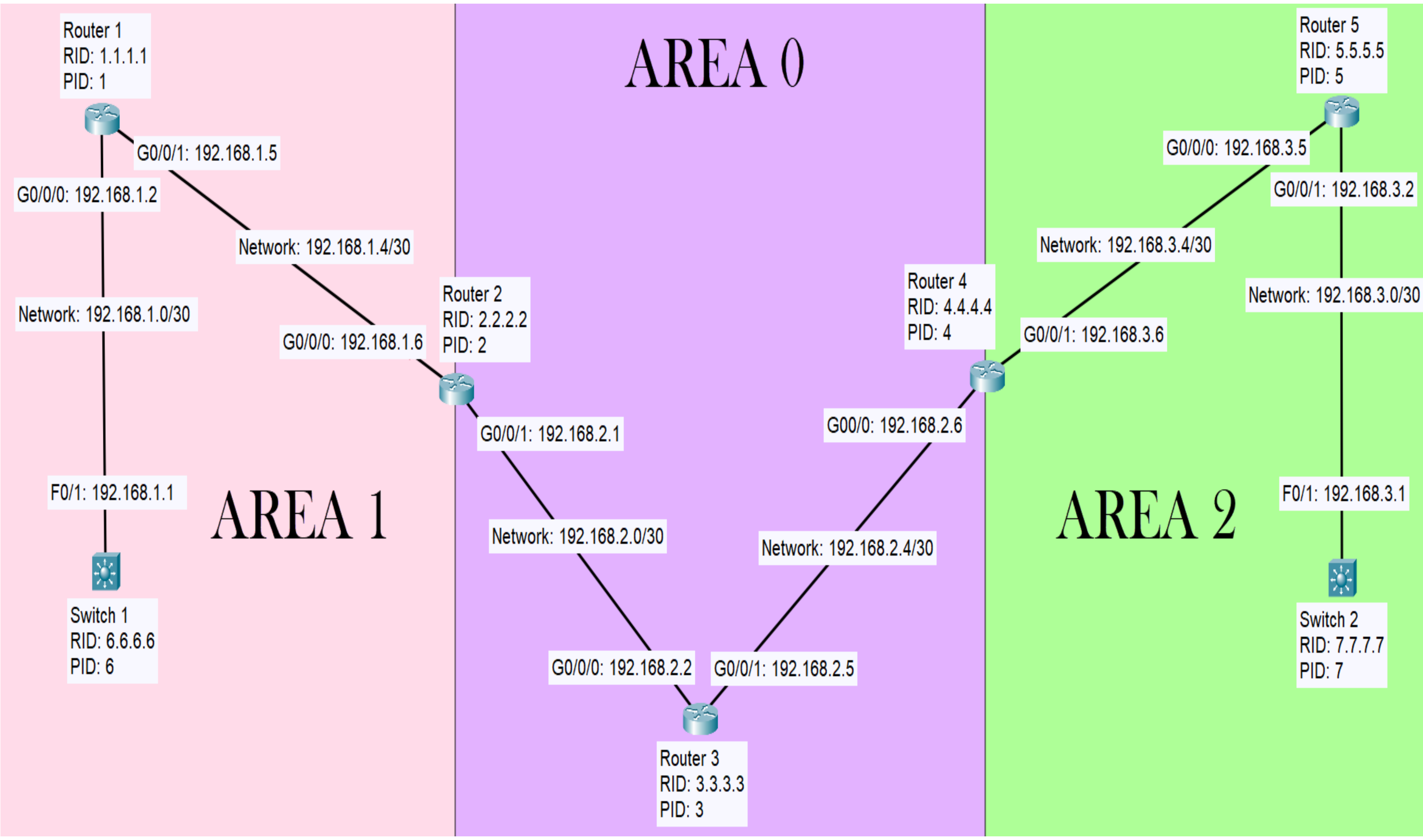
Configures a new OSPFv3 for IPV6 instance that contains all the subsequent OSPFv3 configuration. The area is included in the command, so the network command is obsolete and can't be used however, OSPV3 requires the ipv6-unicast routing command as a prerequisite. The process ID can also be in a range from 1 to 65,535, that uniquely identifies the OSPFv3 process on that local switch and does not need to match the process ID of neighboring routers.

Switch(config-if)#*ip ospf <process-id> area <area-id>*

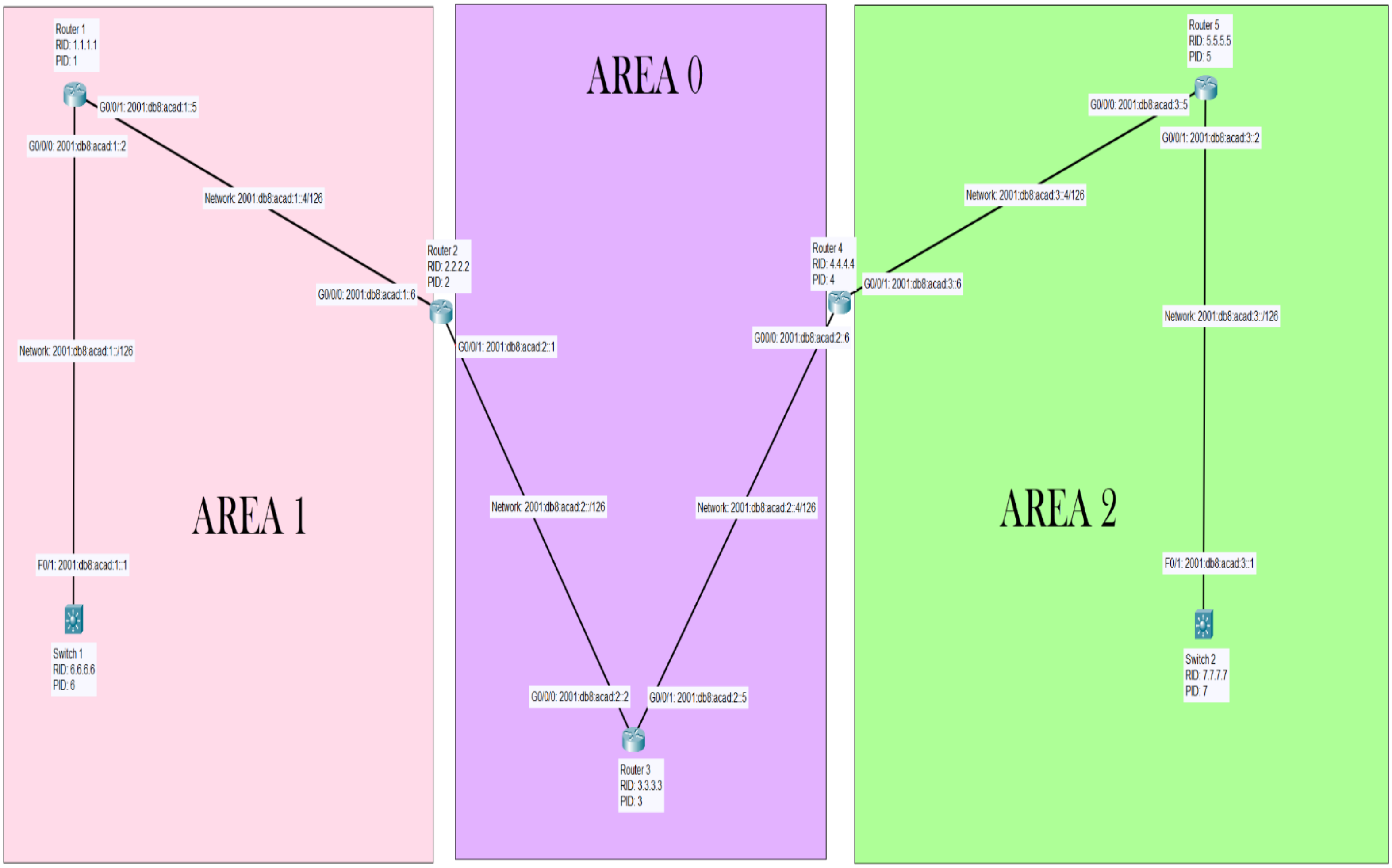
Switch(config-if)#*ipv6 ospf <process-id> area <area-id>*

The *ip ospf* command above enabled OSPFv2 on a specific interface and assigns it to a designated OSPF area while the *ipv6 ospf* command above does the same for OSPFv3 with IPV6. These commands will place a specific interface in an area that you choose to organize and structure the network.

## Network Topology




*Note: The diagram above is a topology for IPV4*




*Note: The diagram above is a topology for IPV6*

## Procedure

```
Area 1:
     Switch1:
     Switch>
     Switch>en
     Switch#conf t
     Switch(config)#ip routing`
     Switch(config)#ipv6 unicast-routing
     Switch(config)#int g1/0/1
     Switch(config-if)#no switchport
     Switch(config-if)#ip add 192.168.1.1 255.255.255.252
     Switch(config-if)#ipv6 add 2001:db8:acad:1::1/126
     Switch(config-if)#no shut
     Switch(config-if)#ip ospf 6 area 1
     Switch(config-if)#ipv6 ospf 6 area 1
     Switch(config-if)#router ospf 6
     Switch(config-router)#network 192.168.1.0
     255.255.255.252 area 1
     Switch(config-router)#router-id 6.6.6.6
     Switch(config-router)#do clear ip ospf process
     Switch(config-router)#exit
     Switch(config-router)#router ospfv3 6
     Switch(config-router)#router-id 6.6.6.6
     Switch(config-router)#exit
     Switch(config)#hostname Switch1

     Router1:
     Router#en
     Router#conf t
     Router(config)#ipv6 unicast-routing
     Router(config)# ip routing
     Router(config)#int g0/0/0
     Router(config-if)#ip address 192.168.1.2
     255.255.255.252
     Router(config-if)#ipv6 address 2001:db8:acad:1::2 /126
     Router(config-if)#no shut
     Router(config-if)#ip ospf 1 area 1
     Router(config-if)#ipv6 ospf 1 area 1
     Router(config-if)#router ospf 1
     Router(config-router)#router-id 1.1.1.1
     Router(config-router)#exit
     Router(config)#int g0/0/1
     Router(config-if)#ip address 192.168.1.5
     255.255.255.252
     Router(config-if)#ipv6 address 2001:db8:acad:1::5/126
     Router(config-if)#no shut
```

```
Router(config-if)#ip ospf 1 area 1
Router(config-if)#ipv6 ospf 1 area 1
Router(config-if)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#exit
Router(config)#hostname Router1

Router2:
Router#en
Router#conf t
Router(config)#ipv6 unicast-routing
Router(config)#ip routing
Router(config)#int g0/0/0
Router(config-if)#ip address 192.168.1.6
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:1::6/126
Router(config-if)#no shut
Router(config-if)#ip ospf 2 area 1
Router(config-if)#ipv6 ospf 2 area 1
Router(config-if)#router ospf 2
Router(config-router)#router-id 2.2.2.2
Router(config-router)#exit
Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.2.1
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:2::1/126
Router(config-if)#no shut
Router(config-if)#ip ospf 2 area 0
Router(config-if)#ipv6 ospf 2 area 0
Router(config-if)#router ospf 2
Router(config-router)#router-id 2.2.2.2
Router(config-router)#exit
Router(config)#hostname Router2
```

Area 0:

```
Router3:
Router#en
Router#conf t
Router(config)#ipv6 unicast-routing
Router(config)#ip routing
Router(config)#int g0/0/0
Router(config-if)#ip address 192.168.2.2
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:2::2/126
Router(config-if)#no shut
Router(config-if)#ip ospf 3 area 0
Router(config-if)#ipv6 ospf 3 area 0
Router(config-if)#router ospf 3
```

```
Router(config-router)#router-id 3.3.3.3
Router(config-router)#exit
Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.2.5
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:2::5/126
Router(config-if)#no shut
Router(config-if)#ip ospf 3 area 0
Router(config-if)#ipv6 ospf 3 area 0
Router(config-if)#router ospf 3
Router(config-router)#router-id 3.3.3.3
Router(config-router)#exit
Router(config)#hostname Router3
```

Area 2:

```
Router4:
Router#en
Router#conf t
Router(config)#ipv6 unicast-routing
Router(config)#ip routing
Router(config)#int g0/0/0
Router(config-if)#ip address 192.168.2.6
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:2::6/126
Router(config-if)#no shut
Router(config-if)#ip ospf 4 area 0
Router(config-if)#ipv6 ospf 4 area 0
Router(config-router)#exit
Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.3.6
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:3::6/126
Router(config-if)#no shut
Router(config-if)#ip ospf 4 area 2
Router(config-if)#ipv6 ospf 4 area 2
Router(config-if)#router ospf 4
Router(config-router)#router-id 4.4.4.4
Router(config-router)#network 192.168.2.4 0.0.0.3 area
0
Router(config-router)#network 192.168.3.4 0.0.0.3 area
2
Router(config-router)#exit
Router(config)#hostname Router4

Router5:
Router#en
Router#conf t
Router(config)#ipv6 unicast-routing
```

```
Router(config)#ip routing
Router(config)#int g0/0/0
Router(config-if)#ip address 192.168.3.5
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:3::5/126
Router(config-if)#no shut
Router(config-if)#ip ospf 5 area 2
Router(config-if)#ipv6 ospf 5 area 2
Router(config-if)#router ospf 5
Router(config-router)#router-id 5.5.5.5
Router(config-router)#exit
Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.3.2
255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:3::2/126
Router(config-if)#no shut
Router(config-if)#ip ospf 5 area 2
Router(config-if)#ipv6 ospf 5 area 2
Router(config-if)#router ospf 5
Router(config-router)#router-id 5.5.5.5
Router(config-router)#exit
Router(config)#hostname Router5
```

```
Switch2:
Switch#en
Switch#conf t
Switch(config)#ip routing
Switch(config)#ipv6 unicast-routing
Switch(config)#int g1/0/1
Switch(config-if)#no switchport
Switch(config-if)#ip address 192.168.3.1
255.255.255.252
Switch(config-if)#no shut
Router(config-if)#ipv6 address 2001:db8:acad:3::1/126
Switch(config-if)#ip ospf 7 area 2
Switch(config-if)#router ospf 7
Switch(config-router)#router-id 7.7.7.7
Switch(config-router)#exit
Switch(config)#int g1/0/1
Switch(config-if)#ipv6 ospf 7 area 2
Switch(config-if)#do clear ip ospf process
Switch(config-if)#exit
Switch(config)#hostname Switch2
```

## Configurations

### Area 1:

Switch1:

```
Switch1(config)#do traceroute 192.168.3.1
Type escape sequence to abort.
Tracing the route to 192.168.3.1
VRF info: (vrf in name/id, vrf out name/id)
1 192.168.1.2 0 msec 4 msec 0 msec
2 192.168.1.6 0 msec 4 msec 0 msec
3 192.168.2.2 0 msec 4 msec 0 msec
4 192.168.2.6 0 msec 4 msec 0 msec
5 192.168.3.5 4 msec 4 msec 0 msec
6 192.168.3.1 4 msec 4 msec *

Switch1(config)#do traceroute 2001:db8:acad:3::1
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD:3::1
1 2001:DB8:ACAD:1::2 0 msec 0 msec 4 msec
2 2001:DB8:ACAD:1::6 0 msec 0 msec 4 msec
3 2001:DB8:ACAD:2::2 0 msec 4 msec 0 msec
4 2001:DB8:ACAD:2::6 0 msec 4 msec 0 msec
5 2001:DB8:ACAD:3::5 0 msec 4 msec 4 msec
6 2001:DB8:ACAD:3::1 0 msec 4 msec 0 msec

Switch1(config)#do show ip ospf
Routing Process "ospf 6" with ID 192.168.1.1
Start time: 00:04:43.184, Time elapsed: 2d21h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msecs
Minimum hold time between two consecutive SPFs 200 msecs
Maximum wait time between two consecutive SPFs 5000 msecs
Incremental-SPF disabled
Initial LSA throttle delay 50 msecs
Minimum hold time for LSA throttle 200 msecs
Maximum wait time for LSA throttle 5000 msecs
Minimum LSA arrival 100 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
```

```
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area 1
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 2d21h ago
SPF algorithm executed 3 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04E6D6
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Switch1(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi1/0/1 6 1 192.168.1.1/30 1 DR 1/1

Switch1(config)#do show ip ospf neighbor detail
Neighbor 1.1.1.1, interface address 192.168.1.2, interface-id 6
In the area 1 via interface GigabitEthernet1/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.1 BDR is 192.168.1.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 2d21h
Index 1/1/1, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Switch1(config)#do show ip int brief
Interface IP-Address OK? Method Status Protocol
```

```
Vlan1 unassigned YES unset administratively down down
GigabitEthernet0/0 unassigned YES unset administratively down
down
GigabitEthernet1/0/1 192.168.1.1 YES manual up up
GigabitEthernet1/0/2 unassigned YES unset down down
GigabitEthernet1/0/3 unassigned YES unset down down
GigabitEthernet1/0/4 unassigned YES unset down down
GigabitEthernet1/0/5 unassigned YES unset down down
GigabitEthernet1/0/6 unassigned YES unset down down
GigabitEthernet1/0/7 unassigned YES unset down down
GigabitEthernet1/0/8 unassigned YES unset down down
GigabitEthernet1/0/9 unassigned YES unset down down
GigabitEthernet1/0/10 unassigned YES unset down down
GigabitEthernet1/0/11 unassigned YES unset down down
GigabitEthernet1/0/12 unassigned YES unset down down
GigabitEthernet1/0/13 unassigned YES unset down down
GigabitEthernet1/0/14 unassigned YES unset down down
GigabitEthernet1/0/15 unassigned YES unset down down
GigabitEthernet1/0/16 unassigned YES unset down down
GigabitEthernet1/0/17 unassigned YES unset down down
GigabitEthernet1/0/18 unassigned YES unset down down
GigabitEthernet1/0/19 unassigned YES unset down down
GigabitEthernet1/0/20 unassigned YES unset down down
GigabitEthernet1/0/21 unassigned YES unset down down
GigabitEthernet1/0/22 unassigned YES unset down down
GigabitEthernet1/0/23 unassigned YES unset down down
GigabitEthernet1/0/24 unassigned YES unset down down
GigabitEthernet1/1/1 unassigned YES unset down down
GigabitEthernet1/1/2 unassigned YES unset down down
GigabitEthernet1/1/3 unassigned YES unset down down
GigabitEthernet1/1/4 unassigned YES unset down down

Switch1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m -
OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       H - NHRP, G - NHRP registered, g - NHRP registration
```

```
summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.1.0/30 is directly connected,
GigabitEthernet1/0/1
L        192.168.1.1/32 is directly connected,
GigabitEthernet1/0/1
O        192.168.1.4/30
           [110/2] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
      192.168.2.0/30 is subnetted, 2 subnets
O IA     192.168.2.0 [110/3] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
O IA     192.168.2.4 [110/4] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
      192.168.3.0/30 is subnetted, 2 subnets
O IA     192.168.3.0 [110/6] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
O IA     192.168.3.4 [110/5] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1

Switch1(config)#do show run
Building configuration...
Current configuration : 9242 bytes

 Last configuration change at 14:20:20 UTC Mon Sep 15 2025

version 17.6
service timestamps debug datetime msec
service timestamps log datetime msec
Call-home is enabled by Smart-Licensing.
service call-home
platform punt-keepalive disable-kernel-core
hostname SWITCH1
vrf definition Mgmt-vrf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
```

```
switch 1 provision c9200l-24t-4g
ip routing
login on-success log
ipv6 unicast-routing
crypto pki trustpoint TP-self-signed-1225728909
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1225728909
revocation-check none
rsakeypair TP-self-signed-1225728909
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
revocation-check crl
crypto pki certificate chain TP-self-signed-1225728909
certificate self-signed 01
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31323235 37323839 3039301E 170D3235 30393132
31373134
35385A17 0D333530 39313231 37313435 385A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
32323537
32383930 39308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 0100CF8D 38F49911 E78379AA 83BB6D2E 6330B072 6E563AE7
EE8B257F
B23F6F0A 65D1B398 12E7EA52 078D3F8C 328EA977 FD3AFA88 35D1B67F
B59C6475
905F987C F491B6BB DA294095 A736477E 03AD81F0 71A16BE0 DC49808D
3C21E80A
40AE4E3D 6B04575B 2EDA6A3B 29F26402 4FB58127 4515C5CA 4F4DDB76
2C1D03D1
A1A9BE67 95C59DD2 D4C2DC96 9EA9E6B3 BDDE81CD A5E85EA6 34CF7E1D
857B1DC0
FFD8C190 03DFADB1 4362DD38 F01B11EE EA6CE877 DE46F38F 0C3B784E
C4869413
66BD828A 2880DF4D 597B7375 E02E4D4D 6678CC4B 4C89C3D7 95668399
7192BCD3
4E2E579D E4425C23 3D7A85E4 02AFC8F5 5630E3B5 7C28C562 5CCF123F
DF0FA98B
90583061 B8FD0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 1476ED6B 8C6D563F A401993B 5807F9C0
FF7F2A6D
```

```
17301D06 03551D0E 04160414 76ED6B8C 6D563FA4 01993B58 07F9C0FF
7F2A6D17
300D0609 2A864886 F70D0101 05050003 82010100 60EB3437 55CAAE90
052AE9E3
7CE968F4 69874583 904D0771 42653C46 3000D8D0 52178233 5E37C7D6
6B8B7931
6CF12405 2FE0BCFC B94DCE8F 157BDD85 058032FD 955E330B 9AAC23BE
17BB584B
FA1B38AA 4B5259C1 3FA935D1 44D9F78D 4501D7F5 E9DE33FA C91726FE
56F5272D
FEB243B9 025B4165 4CCF1E30 3BF6909C 16887FA2 CB00C472 1A5138F2
1C450CD2
6D9A624B 6D66ECF4 2D03F76F 8426AF2A 1BC0E82C EC59BC11 A582E3A5
30534BC9
96B52780 F1AA4CC6 39C58406 9A4619D3 7AFC8DD9 F2F440B3 4844FC7E
AE8456AF
2529F244 DEBAAF70 5773E702 FA06C0E5 8D6739E3 4EF20215 3196360D
320C15C5
DE96DF55 063AAEF4 6AAF6226 5B601309 4C84C1EF
quit
crypto pki certificate chain SLA-TrustPoint
certificate ca 01
30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101
0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317
43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533
30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355
040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73
696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382
010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1
F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388
8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7
D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F
EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF
58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B
42C68BB7
```

```
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8
8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368
95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04
04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604
1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01
010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78
240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2
55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49
1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69
39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD
230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F
719BB2F0
D697DF7F 28
quit
license boot level network-essentials addon dna-essentials
diagnostic bootup level minimal
spanning-tree mode rapid-pvst
spanning-tree extend system-id
memory free low-watermark processor 10633
redundancy
mode sso
transceiver type all
monitoring
class-map match-any system-cpp-police-ewlc-control
description EWLC Control
class-map match-any system-cpp-police-topology-control
description Topology control
class-map match-any system-cpp-police-sw-forward
description Sw forwarding, L2 LVX data packets, LOGGING, Transit
Traffic
class-map match-any system-cpp-default
description EWLC data, Inter FED Traffic
class-map match-any system-cpp-police-sys-data
```

```
description Openflow, Exception, EGR Exception, NFL Sampled
Data, RPF Failed
class-map match-any system-cpp-police-punt-webauth
description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
description L2 LVX control packets
class-map match-any system-cpp-police-forus
description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
description MCAST END STATION
class-map match-any system-cpp-police-high-rate-app
description High Rate Applications
class-map match-any system-cpp-police-multicast
description MCAST Data
class-map match-any system-cpp-police-l2-control
description L2 control
class-map match-any system-cpp-police-dot1x-auth
description DOT1X Auth
class-map match-any system-cpp-police-data
description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
description Stackwise Virtual OOB
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
description DHCP snooping
class-map match-any system-cpp-police-ios-routing
description L2 control, Topology control, Routing control, Low
Latency
class-map match-any system-cpp-police-system-critical
description System Critical and Gold Pkt
class-map match-any system-cpp-police-ios-feature
description
ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTDa
ta,Transit,DOT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusA
RP,McastEndStn,Openflow,Exception,EGRExcption,NflSampled,RpfFail
ed
policy-map system-cpp-policy
interface GigabitEthernet0/0
vrf forwarding Mgmt-vrf
no ip address
shutdown
interface GigabitEthernet1/0/1
no switchport
```

```
ip address 192.168.1.1 255.255.255.252
ip ospf 6 area 1
ipv6 address 2001:DB8:ACAD:1::1/126
ipv6 ospf 6 area 1
interface GigabitEthernet1/0/2
interface GigabitEthernet1/0/3
interface GigabitEthernet1/0/4
interface GigabitEthernet1/0/5
interface GigabitEthernet1/0/6
interface GigabitEthernet1/0/7
interface GigabitEthernet1/0/8
interface GigabitEthernet1/0/9
interface GigabitEthernet1/0/10
interface GigabitEthernet1/0/11
interface GigabitEthernet1/0/12
interface GigabitEthernet1/0/13
interface GigabitEthernet1/0/14
interface GigabitEthernet1/0/15
interface GigabitEthernet1/0/16
interface GigabitEthernet1/0/17
interface GigabitEthernet1/0/18
interface GigabitEthernet1/0/19
interface GigabitEthernet1/0/20
interface GigabitEthernet1/0/21
interface GigabitEthernet1/0/22
interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/24
interface GigabitEthernet1/1/1
interface GigabitEthernet1/1/2
interface GigabitEthernet1/1/3
interface GigabitEthernet1/1/4
interface Vlan1
no ip address
shutdown
router ospfv3 6
router-id 6.6.6.6
address-family ipv6 unicast
exit-address-family
router ospf 6
router-id 6.6.6.6
network 192.168.1.0 0.0.0.3 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
control-plane
service-policy input system-cpp-policy
```

```
line con 0
stopbits 1
line aux 0
line vty 0 4
login
transport input ssh
line vty 5 15
login
transport input ssh
call-home
If contact email address in call-home is configured as sch-
smart-licensing@cisco.com
the email address configured in Cisco Smart License Portal will
be used as contact email address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
profile "CiscoTAC-1"
active
destination transport-method http
end
```

Router1:

```
Router1(config)#do show ip ospf
Routing Process "ospf 1" with ID 1.1.1.1
Start time: 00:51:58.440, Time elapsed: 4d00h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msecs
Minimum hold time between two consecutive SPFs 200 msecs
Maximum wait time between two consecutive SPFs 5000 msecs
Incremental-SPF disabled
Initial LSA throttle delay 50 msecs
Minimum hold time for LSA throttle 200 msecs
Maximum wait time for LSA throttle 5000 msecs
Minimum LSA arrival 100 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
```

```
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area 1
Number of interfaces in this area is 2
Area has no authentication
SPF algorithm last executed 2d22h ago
SPF algorithm executed 58 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04E2D8
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router1(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 1 1 192.168.1.5/30 1 DR 1/1
Gi0/0/0 1 1 192.168.1.2/30 1 BDR 1/1

Router1(config)#do show ip ospf neighbor detail
Neighbor 2.2.2.2, interface address 192.168.1.6, interface-id 6
In the area 1 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.5 BDR is 192.168.1.6
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:37
Neighbor is up for 4d00h
Index 2/2/2, retransmission queue length 0, number of
retransmission 4
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 2
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 192.168.1.1, interface address 192.168.1.1, interface-
id 38
In the area 1 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.1 BDR is 192.168.1.2
Options is 0x12 in Hello (E-bit, L-bit)
```

```
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 2d22h
Index 1/1/1, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Router1(config)#do show ip interface brief
Any interface listed with OK? value "NO" does not have a valid
configuration
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.1.2 YES manual up up
GigabitEthernet0/0/1 192.168.1.5 YES manual up up
Serial0/1/0 unassigned NO unset down down
Serial0/1/1 unassigned NO unset down down
GigabitEthernet0/2/0 unassigned YES unset administratively down
down
GigabitEthernet0/2/1 unassigned YES unset administratively down
down
GigabitEthernet0 unassigned YES unset administratively down down

Router1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 4 subnets, 2 masks
C        192.168.1.0/30 is directly connected,
GigabitEthernet0/0/0
```

```
L        192.168.1.2/32 is directly connected,
GigabitEthernet0/0/0
C        192.168.1.4/30 is directly connected,
GigabitEthernet0/0/1
L        192.168.1.5/32 is directly connected,
GigabitEthernet0/0/1
      192.168.2.0/30 is subnetted, 2 subnets
O IA     192.168.2.0 [110/2] via 192.168.1.6, 3d02h,
GigabitEthernet0/0/1
O IA     192.168.2.4 [110/3] via 192.168.1.6, 2d21h,
GigabitEthernet0/0/1
      192.168.3.0/30 is subnetted, 2 subnets
O IA     192.168.3.0 [110/5] via 192.168.1.6, 02:04:56,
GigabitEthernet0/0/1
O IA     192.168.3.4 [110/4] via 192.168.1.6, 02:05:36,
GigabitEthernet0/0/1

Router1(config)#do show run
Building configuration...
Current configuration : 3990 bytes
Last configuration change at 15:49:23 UTC Thu Sep 11 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-48848711
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-48848711
revocation-check none
rsakeypair TP-self-signed-48848711
crypto pki certificate chain TP-self-signed-48848711
```

certificate self-signed 01

3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
2F312D30 2B060355 04031324 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 34383834 38373131 301E170D 32353039 31313135
34393234
5A170D33 30303130 31303030 3030305A 302F312D 302B0603 55040313
24494F53
2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D343838
34383731
31308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201
01009F2C E8602FF3 1BB9D7BD 0DA227EE E53AD8C4 19534485 C26B93F8
A5392DCF
A5EB64A8 93E9221B AAC5B478 9F99A92F 88448817 045833A2 0E8782F8
509B4146
BA6D420A C82B01F9 C0571795 2A8D5D6B 6F6C2A88 30DA4D73 7BA2DDD8
266D1870
C311C58B 18E2EA65 1CA4460E 36DC46DB AFA5C343 104F3851 A0D325F1
4C76A576
50A1FD5B 73C5D529 EB53C843 31C2F40A 9D1D51A8 D62BA7D4 18FD4F2D
E7499F8A
F9CF2CE9 B2DA7F7D 3DADE7A2 8EDE63CF 7282F31C BD2FE1C8 F7D8D4F3
2D842DFE
4608A8DC F7DBCB80 18C0A205 378A2E5F FD7FB689 642CA2C3 33EA9DAB
43656FD9
9BBF5468 36F1B657 9E3FC51F 91D8BA2A 3BC0602C 25008C08 28BD78AA
5A224834
6BBD0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603
551D2304 18301680 14C7150A D125EB9A 72A307CF 33924973 E073A353
7D301D06
03551D0E 04160414 C7150AD1 25EB9A72 A307CF33 924973E0 73A3537D
300D0609
2A864886 F70D0101 05050003 82010100 1AA8F97C C3FDBF36 588D1D65
EC686BAA
4FAD446C 0086662D 2ABB9EE4 3DBD3D97 5D5AB692 AB5E1DEC D110E877
086879EA
9F6A81EA BC1EA3CA 820E3EED 4BB8E0DE 13671C55 496A8114 4A5A6BA8
1894068D
EBE7D495 F6A00EF4 96B23654 99213BAF 76939456 FA06C1FF 89569438
9012BA6D
9DD03025 20CF31C0 9A689E77 7FD05C6C FB680FB4 8529375E 9CB211C9
8C91F9DB
02DB8066 49DD9971 73823910 FFC8C05C 3608150E 2025B8B8 9D154B78
5B1709D7

```
FF7599EA 0B9C4B2C E2F039B0 F5A2DA97 17CF0569 E2E574EE DEDA63C8
C6FC18BD
4B102CF5 4FDF2E00 D4E500A1 764B8C73 57BEA0E4 7E22DB39 D6851C34
39923117
84B66321 F399D63C F50ED488 CDD65D3A
quit
license udi pid ISR4321/K9 sn FDO214328HZ
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.2 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:1::2/126
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 192.168.1.5 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:1::5/126
ipv6 ospf 1 area 1
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.3 area 1
network 192.168.1.4 0.0.0.3 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
```

```
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router2:

```
Router2(config)#do show ip ospf
Routing Process "ospf 2" with ID 2.2.2.2
Start time: 1w1d, Time elapsed: 4d00h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
It is an area border router
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msecs
Minimum hold time between two consecutive SPFs 200 msecs
Maximum wait time between two consecutive SPFs 5000 msecs
Incremental-SPF disabled
Initial LSA throttle delay 50 msecs
Minimum hold time for LSA throttle 200 msecs
Maximum wait time for LSA throttle 5000 msecs
Minimum LSA arrival 100 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 2. 2 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
```

```
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 01:31:18.433 ago
SPF algorithm executed 55 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04C5D1
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
Area 1
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 2d22h ago
SPF algorithm executed 42 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04DEDA
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router2(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 2 0 192.168.2.1/30 1 BDR 1/1
Gi0/0/0 2 1 192.168.1.6/30 1 BDR 1/1

Router2(config)#do show ip ospf neighbor detail
Neighbor 3.3.3.3, interface address 192.168.2.2, interface-id 6
In the area 0 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.2 BDR is 192.168.2.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:35
Neighbor is up for 3d01h
Index 1/1/2, retransmission queue length 0, number of
retransmission 4
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 1.1.1.1, interface address 192.168.1.5, interface-id 7
```

```
In the area 1 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.5 BDR is 192.168.1.6
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 4d00h
Index 1/1/1, retransmission queue length 0, number of
retransmission 3
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 2, maximum is 2
Last retransmission scan time is 0 msec, maximum is 0 msec

Router2(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.1.6 YES manual up up
GigabitEthernet0/0/1 192.168.2.1 YES manual up up
Serial0/1/0 unassigned YES unset administratively down down
Serial0/1/1 unassigned YES unset administratively down down
GigabitEthernet0 unassigned YES unset administratively down down

Router2(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
O        192.168.1.0/30
           [110/2] via 192.168.1.5, 00:10:52,
GigabitEthernet0/0/0
C        192.168.1.4/30 is directly connected,
```

```
GigabitEthernet0/0/0
L        192.168.1.6/32 is directly connected,
GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.2.0/30 is directly connected,
GigabitEthernet0/0/1
L        192.168.2.1/32 is directly connected,
GigabitEthernet0/0/1
O        192.168.2.4/30 [110/2] via 192.168.2.2, 2d21h,
GigabitEthernet0/0/1
      192.168.3.0/30 is subnetted, 2 subnets
O IA     192.168.3.0 [110/4] via 192.168.2.2, 02:06:03,
GigabitEthernet0/0/1
O IA     192.168.3.4 [110/3] via 192.168.2.2, 02:06:43,
GigabitEthernet0/0/1

Router2(config)#do show run
Building configuration...
Current configuration : 3905 bytes
Last configuration change at 17:01:42 UTC Mon Sep 15 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-1411542050
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1411542050
revocation-check none
rsakeypair TP-self-signed-1411542050
crypto pki certificate chain TP-self-signed-1411542050
certificate self-signed 01
```

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31343131 35343230 3530301E 170D3235 30393131
31343233
30395A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
34313135
34323035 30308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 01009ADC EF88D7AE 1D3307A2 A5AD1B63 2EF8AAF1 533E9A45
009FD5B8
13D7C12F 7A20141D 8B75E6D3 081A8868 7C753E05 28D34C83 51929A33
8DB44D19
14E892B1 34A41170 E499A5CE E804F3CE 4FC26B76 A8CA4C00 48FE969A
2CD0DD93
68FCA989 DE8EAC7C 7EEEC39C EAEC36D5 F3419771 4C7EBBE1 6CAD6483
A610032A
F153A90F 663A9CC0 2CD07B9F 53391363 62E16AC4 9F30BFD6 F24B1DFA
2ED0523D
9D74727A 91B3CE02 CA8B2E1F 80855719 B7EFC54F 6417F72C 2B222A2D
F756AD38
44239C20 93084A42 6FF78C9C D99F84EB D9012FD5 529956A8 839E6C95
AF5A3B0E
3F6A67F7 0E3B4A88 401C898B 15B85941 93248370 3A76B838 9D72B1EF
6BA882FB
297A74D6 B9690203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 14C82A68 7DE5B186 FE679B1C 30F9A7F6
64B3635A
F0301D06 03551D0E 04160414 C82A687D E5B186FE 679B1C30 F9A7F664
B3635AF0
300D0609 2A864886 F70D0101 05050003 82010100 8647BDFC FE685CE2
E3686C2A
CD052A1B A0D2356C 294639A8 6E0B3627 2A7AFF6C 1BE145A5 6AB049DC
43A326CC
475AA583 C5CEA83B 6066994F A7BF408B 5AA6499E 3C9119C8 AE345BEC
9031E689
8DFDA2B1 5F5A30FE 844BF07A 0082D6F3 2B85B989 13774E60 BFC7325F
B95A5B95
CEA03292 8E5CF852 D4CA7ACE B4239E1E CE5C273A F64431C9 2767039A
912EEC15
3E398B74 450FB172 382248B1 1DB6CA89 BE26B357 B9ADC191 C7E8E723
E0173092

```
3CA9E875 FF470610 CB5F2DF2 4732D227 F3DB0A8D 59162E4E 675F54D3
F303770A
B1E4296D 52E04E92 F9FE0E8B 1CE188EF 114740FB 651EB1E2 B45051BD
2A58A3C2
45A36638 DDB29AAE 13915261 58DE0E1A 6823749C
quit
license udi pid ISR4321/K9 sn FDO215009PN
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.6 255.255.255.252
ip ospf 2 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:1::6/126
ipv6 ospf 2 area 1
interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.252
ip ospf 2 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::1/126
ipv6 ospf 2 area 0
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 2
router-id 2.2.2.2
network 192.168.1.4 0.0.0.3 area 1
network 192.168.2.0 0.0.0.3 area 0
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 2
control-plane
line con 0
```

```
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

## Area 0:

### Router3:

```
Router3(config)#do show ip ospf
Routing Process "ospf 3" with ID 3.3.3.3
Start time: 01:13:39.058, Time elapsed: 4d00h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msecs
Minimum hold time between two consecutive SPFs 200 msecs
Maximum wait time between two consecutive SPFs 5000 msecs
Incremental-SPF disabled
Initial LSA throttle delay 50 msecs
Minimum hold time for LSA throttle 200 msecs
Maximum wait time for LSA throttle 5000 msecs
Minimum LSA arrival 100 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
Number of interfaces in this area is 2
Area has no authentication
```

```
SPF algorithm last executed 01:33:53.510 ago
SPF algorithm executed 56 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04C5D1
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router3(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 3 0 192.168.2.5/30 1 BDR 1/1
Gi0/0/0 3 0 192.168.2.2/30 1 DR 1/1

Router3(config)#do show ip ospf neighbor detail
Neighbor 192.168.2.6, interface address 192.168.2.6, interface-
id 6
In the area 0 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.6 BDR is 192.168.2.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 2d20h
Index 1/2/2, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 2.2.2.2, interface address 192.168.2.1, interface-id 7
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.2 BDR is 192.168.2.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:32
Neighbor is up for 3d01h
Index 1/1/1, retransmission queue length 0, number of
retransmission 11
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec

Router3(config)#do show ip interface brief
```

```
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.2.2 YES manual up up
GigabitEthernet0/0/1 192.168.2.5 YES manual up up
Serial0/1/0 unassigned YES unset down down
Serial0/1/1 unassigned YES unset down down
GigabitEthernet0/2/0 unassigned YES unset down down
GigabitEthernet0/2/1 unassigned YES unset down down
GigabitEthernet0 unassigned YES unset down down

Router3(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set

      192.168.1.0/30 is subnetted, 2 subnets
O IA     192.168.1.0 [110/3] via 192.168.2.1, 00:12:37,
GigabitEthernet0/0/0
O IA     192.168.1.4 [110/2] via 192.168.2.1, 3d02h,
GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 4 subnets, 2 masks
C        192.168.2.0/30 is directly connected,
GigabitEthernet0/0/0
L        192.168.2.2/32 is directly connected,
GigabitEthernet0/0/0
C        192.168.2.4/30 is directly connected,
GigabitEthernet0/0/1
L        192.168.2.5/32 is directly connected,
GigabitEthernet0/0/1
      192.168.3.0/30 is subnetted, 2 subnets
O IA     192.168.3.0 [110/3] via 192.168.2.6, 02:07:17,
GigabitEthernet0/0/1
```

```
O IA     192.168.3.4 [110/2] via 192.168.2.6, 02:07:57,
GigabitEthernet0/0/1

Router3(config)#do show run
Building configuration...
Current configuration : 1698 bytes
Last configuration change at 15:26:04 UTC Mon Sep 15 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router3
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214420HS
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.2.2 255.255.255.252
ip ospf 3 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::2/126
ipv6 ospf 3 area 0
interface GigabitEthernet0/0/1
ip address 192.168.2.5 255.255.255.252
ip ospf 3 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::5/126
ipv6 ospf 3 area 0
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
```

```
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router ospf 3
router-id 3.3.3.3
network 192.168.2.0 0.0.0.3 area 0
network 192.168.2.4 0.0.0.3 area 0
ip forward-protocol nd
no ip http server
ip http secure-server
ipv6 router ospf 3
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
End
```

## Area 2:

### Router4:

```
Router4(config)#do show ip ospf
Routing Process "ospf 4" with ID 192.168.2.6
Start time: 00:28:28.630, Time elapsed: 2d21h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
It is an area border router
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msecs
Minimum hold time between two consecutive SPFs 200 msecs
Maximum wait time between two consecutive SPFs 5000 msecs
Incremental-SPF disabled
Initial LSA throttle delay 50 msecs
```

```
Minimum hold time for LSA throttle 200 msecs
Maximum wait time for LSA throttle 5000 msecs
Minimum LSA arrival 100 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 2. 2 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 02:04:40.290 ago
SPF algorithm executed 28 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04B1DB
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
Area 2
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 02:03:44.585 ago
SPF algorithm executed 49 times
Area ranges are
Number of LSA 9. Checksum Sum 0x047BC2
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router4(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/0 4 0 192.168.2.6/30 1 DR 1/1
Gi0/0/1 4 2 192.168.3.6/30 1 DR 1/1
```

```
Router4(config)#do show ip ospf neighbor detail
Neighbor 3.3.3.3, interface address 192.168.2.5, interface-id 7
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.6 BDR is 192.168.2.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:37
Neighbor is up for 2d20h
Index 1/1/1, retransmission queue length 0, number of
retransmission 4
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 5.5.5.5, interface address 192.168.3.5, interface-id 6
In the area 2 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.6 BDR is 192.168.3.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 01:36:29
Index 1/1/2, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Router4(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.2.6 YES manual up up
GigabitEthernet0/0/1 192.168.3.6 YES manual up up
GigabitEthernet0 unassigned YES unset administratively down down

Router4(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
```

```
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set

      192.168.1.0/30 is subnetted, 2 subnets
O IA     192.168.1.0 [110/4] via 192.168.2.5, 00:13:14,
GigabitEthernet0/0/0
O IA     192.168.1.4 [110/3] via 192.168.2.5, 02:08:34,
GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 3 subnets, 2 masks
O        192.168.2.0/30
           [110/2] via 192.168.2.5, 02:08:34,
GigabitEthernet0/0/0
C        192.168.2.4/30 is directly connected,
GigabitEthernet0/0/0
L        192.168.2.6/32 is directly connected,
GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks
O        192.168.3.0/30
           [110/2] via 192.168.3.5, 02:07:48,
GigabitEthernet0/0/1
C        192.168.3.4/30 is directly connected,
GigabitEthernet0/0/1
L        192.168.3.6/32 is directly connected,
GigabitEthernet0/0/1

Router4(config)#do show run
Building configuration...
Current configuration : 3725 bytes
Last configuration change at 20:15:08 UTC Fri Sep 12 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router4
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
```

```
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-187689846
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-187689846
revocation-check none
rsakeypair TP-self-signed-187689846
crypto pki certificate chain TP-self-signed-187689846
certificate self-signed 01
```


3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31383736 38393834 36301E17 0D323530 39313231
39323630
315A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403
1325494F
532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3138
37363839
38343630 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
82010100 C4FFA596 31767A3A 04BB0320 8D4CA352 D8D3811B 7590CDA5
BADB81C6
634CC332 CBC6EE98 8BA62B22 BADA051D 7DB75D82 7E57AE87 87B8CF0A
213A57BC
92D7DCD4 EE7C63D1 02E15906 546C5858 0F90B11F 8045CE69 F451F547
FA090C61
B8C85427 D6EA2733 50BE25C8 302651E7 FA297284 3A41AE84 7ACBB6A3
72146FC0
4AF59E0E 163981F0 96591D27 07B44A37 47617426 6C83BBA9 A782A1B9
9E1B0412
FB809A24 F2809526 CBF14D46 ADA3EAF7 BCDB2BE3 AA230BC3 F2BCB214
22FE8911
D24E1461 7AA70D10 6D6A25D8 4DE942EB 4D87C682 AE8D1F31 ED73892A
67B8C9DB
2B387F3D A867EA5E EBAEAC6E 4453AF5A 3AA99910 4F9ACC16 40135F9B
1FA839D3
C8E0EA7B 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
0603551D 23041830 168014D5 07205C88 D0577F79 9EDE4B39 EE754EC9
E1AAD130

```
1D060355 1D0E0416 0414D507 205C88D0 577F799E DE4B39EE 754EC9E1
AAD1300D
06092A86 4886F70D 01010505 00038201 010014E5 1BDDAD2A A2617015
896ABC0E
0352CC43 8C628996 D28D2539 2B6C1E4A D67D032F D6BBDDF7 1626BA0E
76A17267
52CEA9EB 907A3005 F5027669 740738FD 3C406ED8 689B7F3A 89B11424
E19C348A
27551845 DAFB05BF 05667D2C D7F6E910 8DD54ABF 7B4BD109 1C8E8C51
7BE5188A
F814018F 1C64B003 61880BC3 E2C30590 0FE35C7B C47338C1 0791B974
BB788A6C
D2529CD7 A3755EDA 1DA626E1 D5ECA1F8 720005D2 5043833C 5F403F64
7D7F85B7
2A915CE2 7F520496 58CB4F38 32E3F134 32B3F13F F3E400B1 34453C8C
8F97B58F
2424EC2C D292F07C E0F2D94D DB4723BE D178A846 49F74414 9E4FCB0E
4AC30F6D
F447D758 97BBBBD3 CE6852CA 9E812AB4 204E
quit
license udi pid ISR4321/K9 sn FLM2407014H
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.2.6 255.255.255.252
ip ospf 4 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::6/126
ipv6 ospf 4 area 0
interface GigabitEthernet0/0/1
ip address 192.168.3.6 255.255.255.252
ip ospf 4 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:3::6/126
ipv6 ospf 4 area 2
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 4
router-id 4.4.4.4
network 192.168.2.4 0.0.0.3 area 2
ip forward-protocol nd
```

```
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 4
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router5:

```
Router5(config)#do show ip ospf
Routing Process "ospf 5" with ID 5.5.5.5
Start time: 00:06:12.168, Time elapsed: 02:05:42.612
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msecs
Minimum hold time between two consecutive SPFs 200 msecs
Maximum wait time between two consecutive SPFs 5000 msecs
Incremental-SPF disabled
Initial LSA throttle delay 50 msecs
Minimum hold time for LSA throttle 200 msecs
Maximum wait time for LSA throttle 5000 msecs
Minimum LSA arrival 100 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
```

```
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area 2
Number of interfaces in this area is 2
Area has no authentication
SPF algorithm last executed 02:04:47.661 ago
SPF algorithm executed 9 times
Area ranges are
Number of LSA 9. Checksum Sum 0x047BC2
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router5(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 5 2 192.168.3.2/30 1 BDR 1/1
Gi0/0/0 5 2 192.168.3.5/30 1 BDR 1/1

Router5(config)#do show ip ospf neighbor detail
Neighbor 192.168.3.1, interface address 192.168.3.1, interface-
id 38
In the area 2 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.1 BDR is 192.168.3.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 02:05:44
Index 1/2/2, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 192.168.2.6, interface address 192.168.3.6, interface-
id 7
In the area 2 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.6 BDR is 192.168.3.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 02:05:46
```

```
Index 1/1/1, retransmission queue length 0, number of
retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 2, maximum is 2
Last retransmission scan time is 0 msec, maximum is 0 msec

Router5(config)#do show ip interface brief
Any interface listed with OK? value "NO" does not have a valid
configuration
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.3.5 YES manual up up
GigabitEthernet0/0/1 192.168.3.2 YES manual up up
Serial0/1/0 unassigned NO unset down down
Serial0/1/1 unassigned NO unset down down
GigabitEthernet0 unassigned YES unset administratively down down

Router5(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
ia - IS-IS inter area, * - candidate default, U - per-user
static route
o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from
PfR
Gateway of last resort is not set
192.168.1.0/30 is subnetted, 2 subnets
O IA 192.168.1.0 [110/5] via 192.168.3.6, 00:10:50,
GigabitEthernet0/0/0
O IA 192.168.1.4 [110/4] via 192.168.3.6, 02:05:30,
GigabitEthernet0/0/0
192.168.2.0/30 is subnetted, 2 subnets
O IA 192.168.2.0 [110/3] via 192.168.3.6, 02:05:30,
GigabitEthernet0/0/0
O IA 192.168.2.4 [110/2] via 192.168.3.6, 02:05:30,
GigabitEthernet0/0/0
192.168.3.0/24 is variably subnetted, 4 subnets, 2 masks
C 192.168.3.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.3.2/32 is directly connected, GigabitEthernet0/0/1
C 192.168.3.4/30 is directly connected, GigabitEthernet0/0/0
L 192.168.3.5/32 is directly connected, GigabitEthernet0/0/0
```

```
Router5(config)#do show run
Building configuration...
Current configuration : 3745 bytes
Last configuration change at 17:39:07 UTC Mon Sep 15 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router5
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-1541607431
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1541607431
revocation-check none
rsakeypair TP-self-signed-1541607431
crypto pki certificate chain TP-self-signed-1541607431
certificate self-signed 01
```

```
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31353431 36303734 3331301E 170D3235 30393135
31353339
32385A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
35343136
30373433 31308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 01009F52 4BD39D6D 7E4029DD 3EDCA271 EDD29A5C 5CC2EEE0
5AB308BB
1D061F9A 925DDA4A D4565D1D AA00EF25 4877C9D6 4686A9B3 021FC74D
99D86211
```

```
56E0E14E CB4FB477 9EB9F9DB C49CBDCD FFE99AD5 C42A831A DBB6660F
6653A69D
21A3504D B141E995 5D3AC8AA BAFAE090 885447E7 CFC3A90E 903CDEAE
63C3782E
2D7088B5 47C6D50C 38C55066 FE9361E5 26B0187A CD4003DA A6315CE2
AE959813
ECF38E93 96FB06AA E7E08F5A A25E4E78 191B35EA 4035F053 C4FD8892
C88BBE9D
8E8707E9 4FD1E4B6 5B7695E9 10A4E80C D7B96B7A AFEC92FC 891A946E
FC78574D
0CC6D5C2 5B1427B3 916CCE7D C1DD9F4F 41E950D4 8B05594A 64312D95
73EF3AF4
505F6DEF D1430203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 1464048B 5A670DC1 4F11EE80 759EDFF4
B4ABADB2
32301D06 03551D0E 04160414 64048B5A 670DC14F 11EE8075 9EDFF4B4
ABADB232
300D0609 2A864886 F70D0101 05050003 82010100 3455B7E7 926EA442
3C3194FB
9A130641 2F8CB5A5 2D73DC58 73F0D6D4 197ADB96 B7C660A4 1F5A920D
D2FD64D7
F9C1B0E0 BE1A28B7 C7CAF9EF 1FB6BE85 074B3ACB 06423C6D DCF21F66
4F53AABB
570EE013 00964146 0BD77AE3 54CEAAF0 B321973D 52D43B43 8E8EF664
260C190E
93E2DB21 1B644112 2884CA14 08A1BC46 6F768179 89A6014B 8ED4C0D4
30D16E87
E1CD54FD E0DB4CAE 4742815C B3ECD751 FB303A18 D43EC6C7 6228C1AA
B9DF6E37
242226F0 03D16246 32E22BEA DD54D97B 9F108B1C 9AC3F929 C2F302FA
087217FF
FBACABBC E81128D7 1F009FDF 4B538A08 53B3CADB D106D643 E339FDB9
227E5771
1D06A401 648DF9C9 1E52E7CB 9F1F1DE4 7D78146F
quit
license udi pid ISR4321/K9 sn FDO214420GM
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.3.5 255.255.255.252
ip ospf 5 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:3::5/126
```

```
ipv6 ospf 5 area 2
interface GigabitEthernet0/0/1
ip address 192.168.3.2 255.255.255.252
ip ospf 5 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:3::2/126
ipv6 ospf 5 area 2
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 5
router-id 5.5.5.5
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 5
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Switch2:

```
*Sep 15 16:47:34.882: %SYS-5-CONFIG_I: Configured from console
by console
Switch2>
Switch2>en
Switch2#conf t
Enter configuration commands, one per line. End with CNTL/Z.

Switch2(config)#do traceroute 192.168.1.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
1 192.168.3.2 4 msec 0 msec 4 msec
2 192.168.3.6 0 msec 4 msec 0 msec
3 192.168.2.5 4 msec 4 msec 0 msec
```

```
4 192.168.2.1 0 msec 4 msec 0 msec
5 192.168.1.5 4 msec 4 msec 0 msec
6 192.168.1.1 4 msec 4 msec *

Switch2(config)#do traceroute 2001:db8:acad:1::1
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD:1::1
1 2001:DB8:ACAD:3::2 0 msec 0 msec 4 msec
2 2001:DB8:ACAD:3::6 0 msec 4 msec 0 msec
3 2001:DB8:ACAD:2::5 0 msec 4 msec 0 msec
4 2001:DB8:ACAD:2::1 4 msec 4 msec 0 msec
5 2001:DB8:ACAD:1::5 0 msec 4 msec 0 msec
6 2001:DB8:ACAD:1::1 4 msec 4 msec 0 msec

Switch2(config)#do show ip ospf
Routing Process "ospf 7" with ID 192.168.3.1
Start time: 00:13:22.384, Time elapsed: 2d23h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msecs
Minimum hold time between two consecutive SPFs 200 msecs
Maximum wait time between two consecutive SPFs 5000 msecs
Incremental-SPF disabled
Initial LSA throttle delay 50 msecs
Minimum hold time for LSA throttle 200 msecs
Maximum wait time for LSA throttle 5000 msecs
Minimum LSA arrival 100 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
```

```
Reference bandwidth unit is 100 mbps
Area 2
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 01:45:25.520 ago
SPF algorithm executed 50 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04823F
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Switch2(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi1/0/1 7 2 192.168.3.1/30 1 DR 1/1

Switch2(config)#do show ip ospf neighbor detail
Neighbor 5.5.5.5, interface address 192.168.3.2, interface-id 7
In the area 2 via interface GigabitEthernet1/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.1 BDR is 192.168.3.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 01:46:24
Index 1/1/1, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Switch2(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES unset up down
GigabitEthernet0/0 unassigned YES unset administratively down
down
GigabitEthernet1/0/1 192.168.3.1 YES manual up up
GigabitEthernet1/0/2 unassigned YES unset down down
GigabitEthernet1/0/3 unassigned YES unset down down
GigabitEthernet1/0/4 unassigned YES unset down down
GigabitEthernet1/0/5 unassigned YES unset down down
GigabitEthernet1/0/6 unassigned YES unset down down
GigabitEthernet1/0/7 unassigned YES unset down down
GigabitEthernet1/0/8 unassigned YES unset down down
```

```
GigabitEthernet1/0/9 unassigned YES unset down down
GigabitEthernet1/0/10 unassigned YES unset down down
GigabitEthernet1/0/11 unassigned YES unset down down
GigabitEthernet1/0/12 unassigned YES unset down down
GigabitEthernet1/0/13 unassigned YES unset down down
GigabitEthernet1/0/14 unassigned YES unset down down
GigabitEthernet1/0/15 unassigned YES unset down down
GigabitEthernet1/0/16 unassigned YES unset down down
GigabitEthernet1/0/17 unassigned YES unset down down
GigabitEthernet1/0/18 unassigned YES unset down down
GigabitEthernet1/0/19 unassigned YES unset down down
GigabitEthernet1/0/20 unassigned YES unset down down
GigabitEthernet1/0/21 unassigned YES unset down down
GigabitEthernet1/0/22 unassigned YES unset down down
GigabitEthernet1/0/23 unassigned YES unset down down
GigabitEthernet1/0/24 unassigned YES unset down down
GigabitEthernet1/1/1 unassigned YES unset down down
GigabitEthernet1/1/2 unassigned YES unset down down
GigabitEthernet1/1/3 unassigned YES unset down down
GigabitEthernet1/1/4 unassigned YES unset down down

Switch2(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m -
OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       H - NHRP, G - NHRP registered, g - NHRP registration
summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      192.168.1.0/30 is subnetted, 2 subnets
```

```
O IA     192.168.1.0 [110/6] via 192.168.3.2, 00:31:03,
GigabitEthernet1/0/1
O IA     192.168.1.4 [110/5] via 192.168.3.2, 02:25:42,
GigabitEthernet1/0/1
      192.168.2.0/30 is subnetted, 2 subnets
O IA     192.168.2.0 [110/4] via 192.168.3.2, 02:25:42,
GigabitEthernet1/0/1
O IA     192.168.2.4 [110/3] via 192.168.3.2, 02:25:42,
GigabitEthernet1/0/1
      192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.3.0/30 is directly connected,
GigabitEthernet1/0/1
L        192.168.3.1/32 is directly connected,
GigabitEthernet1/0/1
O        192.168.3.4/30
           [110/2] via 192.168.3.2, 02:25:42,
GigabitEthernet1/0/1

Switch2(config)#do show run
Building configuration...
Current configuration : 9182 bytes
Last configuration change at 14:34:53 UTC Mon Sep 15 2025
version 17.3
service timestamps debug datetime msec
service timestamps log datetime msec
Call-home is enabled by Smart-Licensing.
platform punt-keepalive disable-kernel-core
hostname SWITCH2
vrf definition Mgmt-vrf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
switch 1 provision c9200l-24t-4g
ip routing
login on-success log
ipv6 unicast-routing
no device-tracking logging theft
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
revocation-check crl
crypto pki trustpoint TP-self-signed-733660727
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-733660727
revocation-check none
rsakeypair TP-self-signed-733660727
```

```
crypto pki certificate chain SLA-TrustPoint
certificate ca 01
30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101
0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317
43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533
30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355
040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73
696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382
010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1
F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388
8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7
D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F
EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF
58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B
42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8
8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368
95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04
04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604
1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01
010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78
240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2
55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49
1765308B
```

```
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69
39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD
230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F
719BB2F0
D697DF7F 28
quit
crypto pki certificate chain TP-self-signed-733660727
certificate self-signed 01
3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 37333336 36303732 37301E17 0D323530 39313231
36313731
375A170D 33353039 31323136 31373137 5A303031 2E302C06 03550403
1325494F
532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3733
33363630
37323730 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
82010100 B854E376 16159541 0DC4254C 264A0048 EEF33ED8 D86CD3BF
0E6E7318
F9CAA299 1F6A70BE 0DC72B46 C874C937 CDB8B682 2578E815 3099FC7D
B0878F40
BD518162 09EF524C 33C54939 E66B93CB 4336C9F4 A210982D 02E66A18
BF0BAB77
EF02DC92 C95B6AD7 7E21F392 9A7C9B8C 2D9F5A15 10A897A9 49B4A52C
806698BB
6948C1E4 9E9C2366 593F392B 5AAFB9BD 98F54F2C EDBC5CFC 0FC58C99
4B82E966
AB72FB0C A1ACE3C4 2841A419 0C293403 578D2315 6AE2E950 C9394D83
36D7548F
95F44C6B D34526C0 2FD1F328 492E6E1D EC3FE764 292AD48B 46323DDD
E8E5B65A
443B8607 65AB8D51 AA12FFB9 4A54ACF2 9A832775 DE7E514C 1B05E286
A432D1D8
1A918023 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
0603551D 23041830 16801471 E8B29DA9 F4C6FCA9 53AB647A 4D899C70
F6C9CC30
1D060355 1D0E0416 041471E8 B29DA9F4 C6FCA953 AB647A4D 899C70F6
C9CC300D
06092A86 4886F70D 01010505 00038201 01002CCC DEDA8DB9 E5EE7F34
09B33AC4
```

3B69CB23 B187D111 AEADBF80 13502C0E 477FC370 E76587C1 9BC5109E
DFEA589F
32C1A677 0CFBB2E0 F6414D62 092E1D9B 66E73478 167726F9 097AA619
946F9B23
D323FF99 E60165FE 7CCC3F8D F9A1AE93 13DBA2CF E97FB766 C0DBA947
A0576A61
425E3392 E8E819D9 9C8D5409 20FCF85A AD8D6720 ADACE366 D9BCA837
C7FE743D
B9048EFF 8F5DCBBA A3131540 317FD540 EF83454F B715AEE0 F4B49C60
0F47E7A0
511458B6 35DF99EA A772D75F 25B22AFC 5B8C848E AF08910A 53BB2333
8753A36A
34E83F10 2BE30F8A B17EBC8D 6FF36010 0D581945 DF5E68A7 044DD9E7
C85CB893
53848ACF FDFD226A AC83C7D0 4584EDCA B852

```
quit
license boot level network-essentials addon dna-essentials
diagnostic bootup level minimal
spanning-tree mode rapid-pvst
spanning-tree extend system-id
memory free low-watermark processor 10308
redundancy
mode sso
transceiver type all
monitoring
class-map match-any system-cpp-police-ewlc-control
description EWLC Control
class-map match-any system-cpp-police-topology-control
description Topology control
class-map match-any system-cpp-police-sw-forward
description Sw forwarding, L2 LVX data packets, LOGGING, Transit
Traffic
class-map match-any system-cpp-default
description EWLC data, Inter FED Traffic
class-map match-any system-cpp-police-sys-data
description Openflow, Exception, EGR Exception, NFL Sampled
Data, RPF Failed
class-map match-any system-cpp-police-punt-webauth
description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
description L2 LVX control packets
class-map match-any system-cpp-police-forus
description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
description MCAST END STATION
class-map match-any system-cpp-police-high-rate-app
description High Rate Applications
```

```
class-map match-any system-cpp-police-multicast
description MCAST Data
class-map match-any system-cpp-police-l2-control
description L2 control
class-map match-any system-cpp-police-dot1x-auth
description DOT1X Auth
class-map match-any system-cpp-police-data
description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
description Stackwise Virtual OOB
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
description DHCP snooping
class-map match-any system-cpp-police-ios-routing
description L2 control, Topology control, Routing control, Low
Latency
class-map match-any system-cpp-police-system-critical
description System Critical and Gold Pkt
class-map match-any system-cpp-police-ios-feature
description
ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTDa
ta,Transit,DOT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusA
RP,McastEndStn,Openflow,Exception,EGRExcption,NflSampled,RpfFail
ed
policy-map system-cpp-policy
interface GigabitEthernet0/0
vrf forwarding Mgmt-vrf
no ip address
shutdown
negotiation auto
interface GigabitEthernet1/0/1
no switchport
ip address 192.168.3.1 255.255.255.252
ip ospf 7 area 2
ipv6 address 2001:DB8:ACAD:3::1/126
ipv6 ospf 7 area 2
interface GigabitEthernet1/0/2
interface GigabitEthernet1/0/3
interface GigabitEthernet1/0/4
interface GigabitEthernet1/0/5
interface GigabitEthernet1/0/6
interface GigabitEthernet1/0/7
interface GigabitEthernet1/0/8
```

```
interface GigabitEthernet1/0/9
interface GigabitEthernet1/0/10
interface GigabitEthernet1/0/11
interface GigabitEthernet1/0/12
interface GigabitEthernet1/0/13
interface GigabitEthernet1/0/14
interface GigabitEthernet1/0/15
interface GigabitEthernet1/0/16
interface GigabitEthernet1/0/17
interface GigabitEthernet1/0/18
interface GigabitEthernet1/0/19
interface GigabitEthernet1/0/20
interface GigabitEthernet1/0/21
interface GigabitEthernet1/0/22
interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/24
interface GigabitEthernet1/1/1
interface GigabitEthernet1/1/2
interface GigabitEthernet1/1/3
interface GigabitEthernet1/1/4
interface Vlan1
no ip address
router ospf 7
router-id 7.7.7.7
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ipv6 router ospf 7
control-plane
service-policy input system-cpp-policy
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
transport input ssh
line vty 5 15
login
transport input ssh
call-home
If contact email address in call-home is configured as sch-
smart-licensing@cisco.com
the email address configured in Cisco Smart License Portal will
be used as contact email address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
```

```
profile "CiscoTAC-1"
active
destination transport-method http
end
```

## Problems

There were 3 main problems that came up in the process of configuring this network, the versions of the routers were not the same, the Multi-Layer Switches had a different hardware so we needed to replace the device with a newer Multi-Layer Switch, and the Ethernet Cables were not properly terminated and caused certain interfaces to not be able to locate each other or turn on. For the versions of the routers, out of the 5 routers used originally 2 routers were on version 15.5 which is not compatible with the other routers that had version 16.9. This significantly delayed the progress of the configuration and caused a delay, taking up time to bring in new routers with updated versions to replace the older routers. For the Multi-Layer Switch, the hardware for each was significantly different from each other causing a delay time to find a replacement for 1 of the Multi-Layer Switches. The Final problem of the Ethernet Cables not working were simply solved by terminating our own Ethernet Cables with an RJ-45 Connector and replacing the faulty cables.

## Conclusion

In this analysis/guide, I set up a Multi Area OSPFv3 focused network including devices like Routers and Multi-Layer Switches. There are also various parts in the analysis where I explained the basic concepts of OSPF, Link State Advertisements, OSPFv2 vs. OSPFv3, Multi-Layer Switches vs. Layer 2 Switches and Single Areas vs. Multi Areas. A list of commands needed to configure the network as shown in this guide is available and pictures of the topology for IPV4 and IPV6 are included. This analysis is enough for any personnel and specifically network administrators who want to configure OSPF to make their network more scalable, efficient and organized while cutting power/bandwidth costs.